

Sound Wave Propagation in a Cylindrical Pipe Subjected to a Linearly Increasing Temperature Profile

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1. Introduction

This project relates to an experiment in acoustics. The selected study investigates the behavior of sound waves traveling in a cylindrical pipe subjected to a temperature profile using numerical methods, with sound wave propagation produced by pistons at the ends of the pipe moving harmonically in the x -direction. A system of two coupled linear hyperbolic first order partial differential equations is derived, which is equivalent to the one-dimensional second-order wave equation. The assumptions used in the derivation process confirm the system can be used for variable speed of sound. A Lax-Wendroff one-step scheme is derived to solve the system of equations, with consistency, order of accuracy, and stability confirmed by analyses. Several scenarios are presented with results that confirm the practicality of the scheme for the case of a constant or linearly increasing temperature profile in the pipe.

2. The Wave Equations for Acoustics

The standard 1-D wave equation in acoustics has the following typical form:

$$\frac{\partial^2 P'}{\partial t^2} = c_o^2 \frac{\partial^2 P'}{\partial x^2} \quad (1)$$

where P' is the perturbation pressure, c_o is the speed of the traveling perturbation pressure, t is time, and x is the position in the space dimension. (Nomenclature is included in Appendix I.) For this project a variable temperature profile in the pipe is eventually used. To keep things manageable, the temperature profile was selected to be linear and increasing with increasing x . Equation 1 is provided in numerous sources, but often the assumptions made in the derivation are not always clear. As we know, the speed of sound in air is a function of temperature alone, assuming air is treated as an ideal gas. It was not clear in Eq. 1 whether c_o was assumed to be constant. Therefore, a preliminary step was taken to derive a representative wave equation for which the speed of sound is not assumed to be constant.

Wave equation derivation assumptions:

- The fluid in the cylinder is air, modeled as an ideal gas.
- The air has values of unperturbed pressure, temperature, and density, which are time independent and denoted by P_o , T_o , and ρ_o . In this study, P_o is also x -direction independent, and is therefore a constant. T_o and ρ_o vary in the x -direction according to the ideal gas law equation of state: $P_o = \rho_o R T_o = \text{constant}$, where R is the ideal gas constant for air.
- The velocity of the unperturbed air in the cylinder is zero.
- The air is frictionless (viscosity is zero).
- There is no heat transfer between parts of the wave having different perturbed temperatures (adiabatic process); furthermore, there are no irreversibilities and so the process is isentropic.
- The body forces are negligible.
- The cylinder axis is parallel to wave propagation.
- Pipe wall is rigid, smooth, and adiabatic.

Conservation of mass, momentum, and energy equations yield the following system of quasi-linear first order partial differential equations (PDE) [1]:

$$\rho_t + \nabla \cdot (\rho \vec{v}) = 0 \quad (2)$$

$$\rho \vec{v}_t + \rho (\vec{v} \cdot \nabla) \vec{v} + \nabla P = 0 \quad (3)$$

$$P_t + \vec{v} \cdot \nabla P - a^2 (\rho_t + \vec{v} \cdot \nabla \rho) = 0 \quad (4)$$

here $\vec{v}$ is the velocity and a the speed of sound. In 1-D, the above equations become:

$$\rho_t + \rho u_x + u \rho_x = 0 \quad (5)$$

$$\rho u_t + \rho u u_x + P_x = 0 \quad (6)$$

$$P_t + u P_x - a^2 (\rho_t + u \rho_x) = 0 \quad (7)$$

where u is the perturbation velocity of air molecules as the wave travels across a point, and not to be confused with the wave velocity. Combining Eq. 5 and 7 to remove the partial derivatives of ρ , and noting that $u = u_o + u' = u'$, $P = P_o + P'$, $\rho = \rho_o + \rho'$, and $a = a_o + a'$, yields:

$$(\rho_o + \rho') u'_t + (\rho_o + \rho') u' u'_x + \frac{\partial}{\partial x} (P_o + P') = 0 \quad (8)$$

$$\frac{\partial}{\partial t} (P_o + P') + u' \frac{\partial}{\partial x} (P_o + P') + a^2 (\rho_o + \rho') u'_x = 0 \quad (9)$$

Eliminating the negligible second order perturbation terms, and noting that $\rho_o a_o^2 = \gamma P_o$ results in the following system of first order linear PDEs:

$$u'_t + \frac{1}{\rho_o} P'_x = 0 \quad (10)$$

$$P'_t + \rho_o a_o^2 u'_x = 0 \quad (11)$$

There have been no assumptions made to this point suggesting that ρ_o or a_o need to be constant. In other words, ρ_o and a_o can vary with x in general. It proves convenient shortly to combine the equation for the speed of sound in an ideal gas, $a_o = \sqrt{\gamma R T_o}$, with the ideal gas equation of state, $P_o = \rho_o R T_o$, and revise Eq. 11 to the following:

$$P'_t + \gamma P_o u'_x = 0 \quad (12)$$

where γ is the specific heats ratio and assumed to be constant and equal to 1.4.

From this point forward, the perturbations u' and P' will be simply denoted as u and P . Second order PDEs for u and P can be derived by taking partial derivatives of Eq. 10 and 12 and combining the results. For u , we can take the partial derivative w.r.t. t of Eq. 10 and the partial derivative w.r.t. x of Eq. 11 and combine the result to obtain:

$$\frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2} \quad (13)$$

Similarly, for P , we can take the partial derivative w.r.t. x of Eq. 10 and the partial derivative w.r.t. t of Eq. 11 and combine the result to obtain:

$$\frac{\partial^2 P}{\partial t^2} = \frac{\partial}{\partial x} \left(a^2 \frac{\partial P}{\partial x} \right) \quad (14)$$

3. The Lax-Wendroff Scheme

Numerical solution of the system of first order PDEs represented by Eq. 10 and 12 in this study is based on the Lax-Wendroff method. This method is a finite difference explicit second order in space and time scheme. It can be derived using a Taylor's series expansion in time up to the second derivative. For u , the finite difference formula is

$$u_j^{n+1} = u_j^n + \frac{\partial u_j^n}{\partial t} k + \frac{1}{2} \frac{\partial^2 u_j^n}{\partial t^2} k^2 + O(k^3) \quad (15)$$

where k is the time step size. The two time derivatives are found using Eq. 10 and 13:

$$\frac{\partial u_j^n}{\partial t} = -\frac{1}{\rho_{oj}} \frac{\partial P_j^n}{\partial x} \quad (16)$$

$$\frac{\partial^2 u_j^n}{\partial t^2} = a_{oj}^2 \frac{\partial^2 u_j^n}{\partial x^2} \quad (17)$$

Substituting Eq. 16 and 17 into 15 yields:

$$u_j^{n+1} = u_j^n - \frac{k}{\rho_{oj}} \frac{\partial P_j^n}{\partial x} + \frac{a_{oj}^2 k^2}{2} \frac{\partial^2 u_j^n}{\partial x^2} + O(k^3) \quad (18)$$

Approximating the two space derivatives with second order centered-difference approximations gives the Lax-Wendroff equation for u :

$$u_j^{n+1} = u_j^n + -\frac{k}{2\rho_{oj}h} (P_{j+1}^n - P_{j-1}^n) + \frac{a_{oj}^2 k^2}{2h^2} (u_{j-1}^n - 2u_j^n + u_{j+1}^n) \quad (19)$$

where h is the space step size. Similarly, for P the finite difference formula is

$$P_j^{n+1} = P_j^n + \frac{\partial P_j^n}{\partial t} k + \frac{1}{2} \frac{\partial^2 P_j^n}{\partial t^2} k^2 + O(k^3) \quad (20)$$

The two time derivatives are found using Eq. 11 and 14:

$$\frac{\partial P_j^n}{\partial t} = -\rho_{oj} a_{oj}^2 \frac{\partial u_j^n}{\partial x} \quad (21)$$

$$\frac{\partial^2 P_j^n}{\partial t^2} = \frac{\partial}{\partial x} \left(a_{oj}^2 \frac{\partial P_j^n}{\partial x} \right) = \gamma P_o \left(\frac{1}{\rho_o} \frac{\partial^2 P_j^n}{\partial x^2} + \frac{\partial P_j^n}{\partial x} \frac{\partial (1/\rho_o)}{\partial x} \right) \quad (22)$$

Substituting Eq. 21 and 22 into 20 yields:

$$P_j^{n+1} = P_j^n - \rho_{oj} a_{oj}^2 \frac{\partial u_j^n}{\partial x} k + \frac{1}{2} \gamma P_o \left(\frac{1}{\rho} \frac{\partial^2 P_j^n}{\partial x^2} + \frac{\partial P_j^n}{\partial x} \frac{\partial (1/\rho_o)}{\partial x} \right) k^2 + O(k^3) \quad (23)$$

Approximating the two space derivatives with second order centered-difference approximations gives the Lax-Wendroff equation for P :

$$P_j^{n+1} = P_j^n - \frac{\gamma P_o k}{2h} (u_{j+1}^n - u_{j-1}^n) + \frac{a_{oj}^2 k^2}{2h^2} (P_{j-1}^n - 2P_j^n + P_{j+1}^n) + \frac{\gamma P_o k^2}{8h^2} (P_{j+1}^n - P_{j-1}^n) ((1/\rho_o)_{j+1}^n - (1/\rho_o)_{j-1}^n) \quad (24)$$

In several tested scenarios, the last term in Eq. 24 is small when compared to the other terms on the RHS of the equal sign, and was considered negligible.

4. Accuracy Analysis of the Lax-Wendroff Scheme

An accuracy analysis was performed for Eq. 19 and 24, resulting in modified difference equations which suggest that the scheme is consistent and has accuracy of second order in space and time:

$$u_t + \frac{1}{\rho_o} P_x = \frac{1}{6} \left(\frac{a_o^2}{\rho_o} k^2 - \frac{1}{\rho_o} h^2 \right) P_{xxx} + \text{higher order terms} \quad (25)$$

$$P_t + \gamma P_o u_x = \frac{k^2}{6} \left(\frac{\gamma^2 P_o^2}{\rho_o} u_{xxx} + \gamma^2 P_o^2 \frac{\partial(1/\rho_o)}{\partial x} u_{xx} \right) - \frac{1}{6} \gamma P_o h^2 u_{xxx} + \text{higher order terms} \quad (26)$$

5. Stability Analysis of the Lax-Wendroff Scheme

A stability analysis was performed on the scheme using Fourier analysis. The system of equations in matrix form are:

$$\begin{bmatrix} \hat{u}^{n+1} \\ \hat{p}^{n+1} \end{bmatrix} = G \begin{bmatrix} \hat{u}^n \\ \hat{p}^n \end{bmatrix} \quad (27)$$

where G is the amplification matrix given by:

$$G = \begin{bmatrix} 1 + \frac{a_o^2 k^2}{h^2} (\cos \xi - 1) & -\frac{k}{\rho_o h} i \sin \xi \\ -\frac{\gamma P_o k}{h} i \sin \xi & 1 + \frac{a_o^2 k^2}{h^2} (\cos \xi - 1) + \frac{\gamma P_o k^2}{2h} \frac{\partial(\frac{1}{\rho_o})}{\partial x} i \sin \xi \end{bmatrix} \quad (28)$$

If the fourth term in $G_{2,2}$, corresponding to the last term in Eq. 24, is neglected, which is a safe assumption, then solving for the eigenvalues of G suggests that stability is guaranteed if $a_o k/h \leq 1$ [1]. Note that in the case of variable a_o with constant space step h , the maximum allowable value of k to maintain stability is restricted to where a_o is a maximum. This restriction implies that $a_o k/h = 1$ where a_o is a maximum and $a_o k/h < 1$ everywhere else.

6. Initial and Boundary Conditions

A schematic of the pipe together with basic assumptions and boundary conditions is shown below:

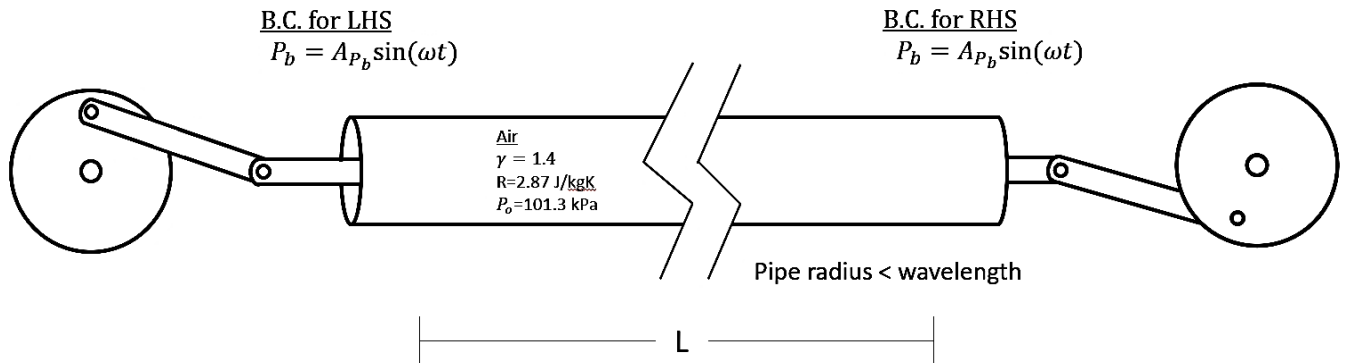


Fig 1. Schematic of pipe showing pistons used to generate boundary conditions.

Solution of the system of first order PDEs for the scheme require a boundary condition at each end of the pipe as well as initial conditions. The focus of this study was on including harmonically changing P and u at the boundaries, with initial conditions for perturbation pressure and velocity set to zero. The pressure perturbations generated by the pistons translate to harmonic pressure waves traveling in the x -direction. The vibrations are assumed to occur at frequencies low enough that the waves are effectively planar.

Phantom nodes were set up for the mesh and located outside the boundary of the system. The boundaries of the system are assumed to be the furthest points towards the ends of the pipe (and inside the pipe) to which the piston faces do not extend beyond.

Boundary perturbation pressure is calculated as

$$P_b = A_{P_b} \sin(\omega t) \quad (29)$$

where A_{P_b} is the amplitude of the harmonic perturbation pressure and ω is the harmonic frequency.

Perturbation pressure at the phantom node at either end of the pipe was calculated using the second order boundary condition formula, for example at the LHS: $P_1^n = 2 P_b^n - P_2^n$.

The phantom node value for u was calculated using:

$$\frac{\partial u}{\partial x} = -\frac{1}{\gamma P_o} \frac{\partial P_b}{\partial t} \quad (30)$$

where $\partial P_b / \partial t$ was found by taking the derivative of Eq. 29 w.r.t. time:

$$\frac{\partial P_b}{\partial t} = A_{P_b} \omega \cos(\omega t) \quad (31)$$

7. Simulation Results

Several simulations were run with different parameters and boundary conditions. Results for a subset of the cases are summarized in the following sections. MATLAB programs are included in Appendix II.

7.1 Fixed space step (h) and fixed T_o

The first simulation was performed with constant T_o to test the accuracy of the scheme with increasing number of nodes. A summary of the simulation parameters is shown in Table 1, followed by two plots showing the results for the perturbation pressure profile. The first plot shows the advancing waves from both boundaries just before they reach the middle of the pipe at $t=0.12s$. The second plot shows a pressure distribution comparison for variable number of nodes with duration set to just before the initial wave from either boundary reaches the opposite boundary.

Table 1. Parameters for accuracy test with fixed space step (h) and fixed T_o .

Parameter	Units	Case	
		1 (Fig 2)	2 (Fig 3)
Pipe length (L)	m	100	100
Number of nodes (incl. phantom) (N)	d-less	200	50, 100, 200, 400
x-Dimension spacing (h)	m	0.505	2.08, 1.02, 0.505, 0.251
Time step (k)	s	0.0015	0.006, 0.0029, 0.0015, 0.000724
$a_o k / h$	d-less	1.0	1.0
Duration	s	0.12	0.25
T_o	K	300	300
<i>Boundary condition LHS</i>			
Perturbation pressure amplitude (A_{P_b})	Pa	1000	1000
Frequency (f)	Hz	10	10

Boundary condition RHS

Perturbation pressure amplitude (A_{P_b})	Pa	300	300
Frequency (f)	Hz	17	17

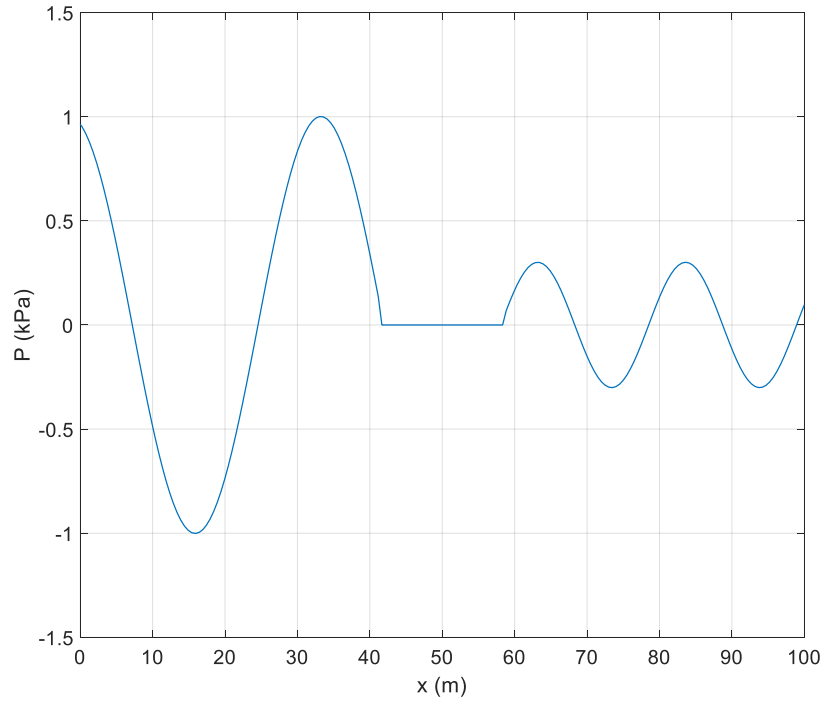


Fig 2. Advancing waves from boundaries for fixed space step (h) and fixed T_o for 0.12s duration.

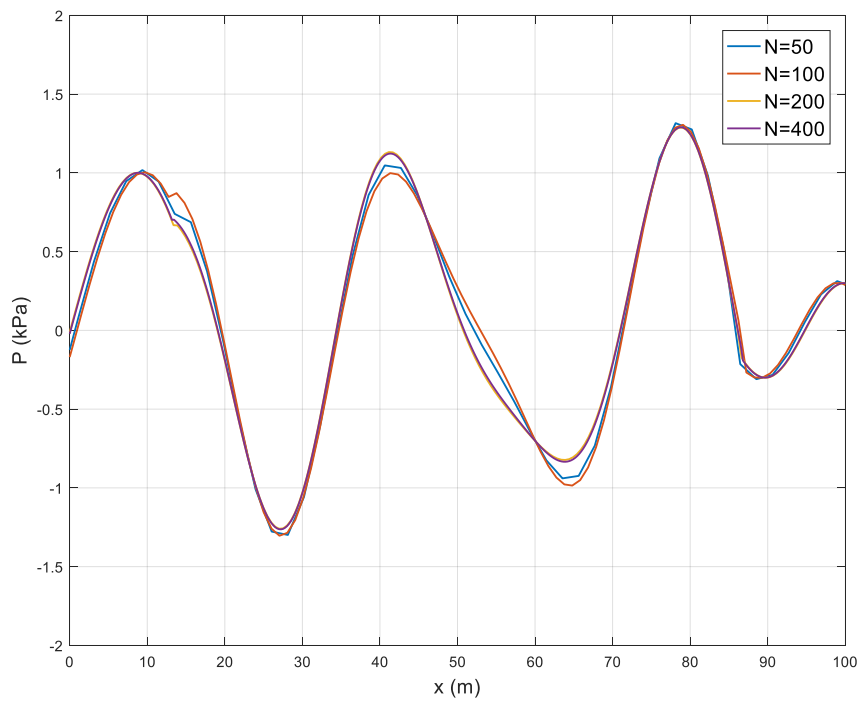


Fig 3. Pressure distribution comparison for variable number of nodes (N) for fixed space step (h) and fixed T_o for 0.25s duration.

Figure 3 suggests that N=200 is a reasonable mesh size for the chosen parameters given that the curves for N=200 and N=400 overlap closely. Note that if the frequency of the vibrating boundary perturbation pressure is increased significantly, the mesh size would likely need to be finer.

7.2 Fixed space step (h) and increasing T_o

The characteristic equation for the case of increasing T_o in the x -direction is given by

$$\frac{dx}{dt} = \pm a_o = \pm \sqrt{\gamma R T_o} = \pm \sqrt{\gamma R \left(T_L + \frac{T_R - T_L}{L} x \right)} \quad (32)$$

The characteristic trajectories are not straight lines as in the case of constant a_o ; they are illustrated in the following plot:

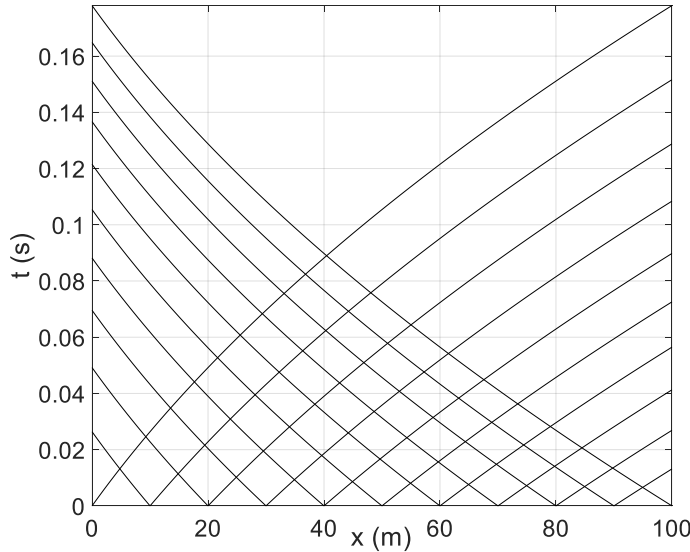


Fig 4. Characteristics trajectories for the case of increasing T_o in the x -direction ($T_o = T_L + \frac{T_R - T_L}{L} x$).

A simulation was performed with linearly increasing T_o (in the x -direction). Simulation parameters are shown in Table 2. The first plot shows the character of an advancing wave (emanating from the LHS boundary) as the speed of sound increases. The advancing wave reaches the RHS boundary of the pipe at approximately $t=0.17$ s. The second series of plots show pressure distribution at incremental time slices for the case with waves emanating from both boundaries.

Table 2. Parameters for fixed space step (h) and linearly increasing T_o in the x -direction.

Parameter	Units	Case	
		3 (Fig 5)	4 (Fig 6)
Pipe length (L)	m	100	100
Number of nodes (incl. phantom) (N)	d-less	500	500
x-Dimension spacing (h)	m	0.201	0.201
Time step (k)	s	0.000259	0.000259
$a_o k/h$	d-less	max of 1.0	max of 1.0

Duration	s	0.17	0.5
T_o	K	300 at LHS, linear to 1500 at RHS	300 at LHS, linear to 1500 at RHS
<i>Boundary condition LHS</i>			
Perturbation pressure amplitude (A_{P_b})	Pa	1000	1000
Frequency (f)	Hz	40	40
<i>Boundary condition RHS</i>			
Perturbation pressure amplitude (A_{P_b})	Pa	-	500
Frequency (f)	Hz	-	23

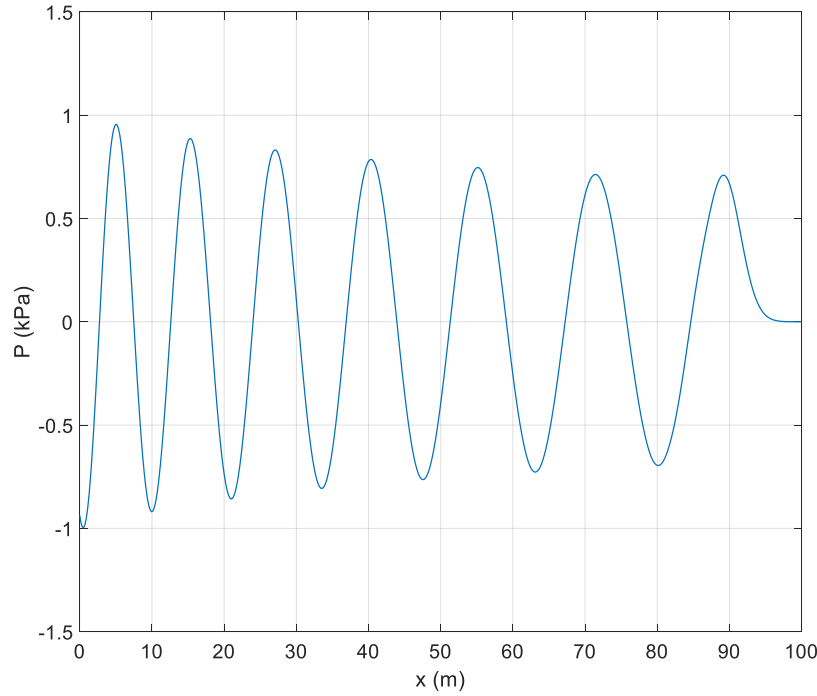


Fig 5. Pressure distribution for fixed space step (h) and increasing T_o for 0.17s duration; wave emanates from LHS boundary.

Figure 5 illustrates how the wavelength increases with increasing speed of sound in the x -direction, as expected. $a_o k/h$ varies from 0.45 to 1.0 because the speed of sound varies in the x -direction. Therefore, we expect third order dispersion error in the results. The time step was selected so that the maximum value of $a_o k/h$ is 1.0 to guarantee stability. Also, apparent in Fig. 5 is the decreasing wave amplitude in the x -direction. It is not clear if this is a physical phenomenon or an artifact of the third order dispersion error in the scheme. The decreasing wave amplitude did not change when a much finer mesh size was tested.

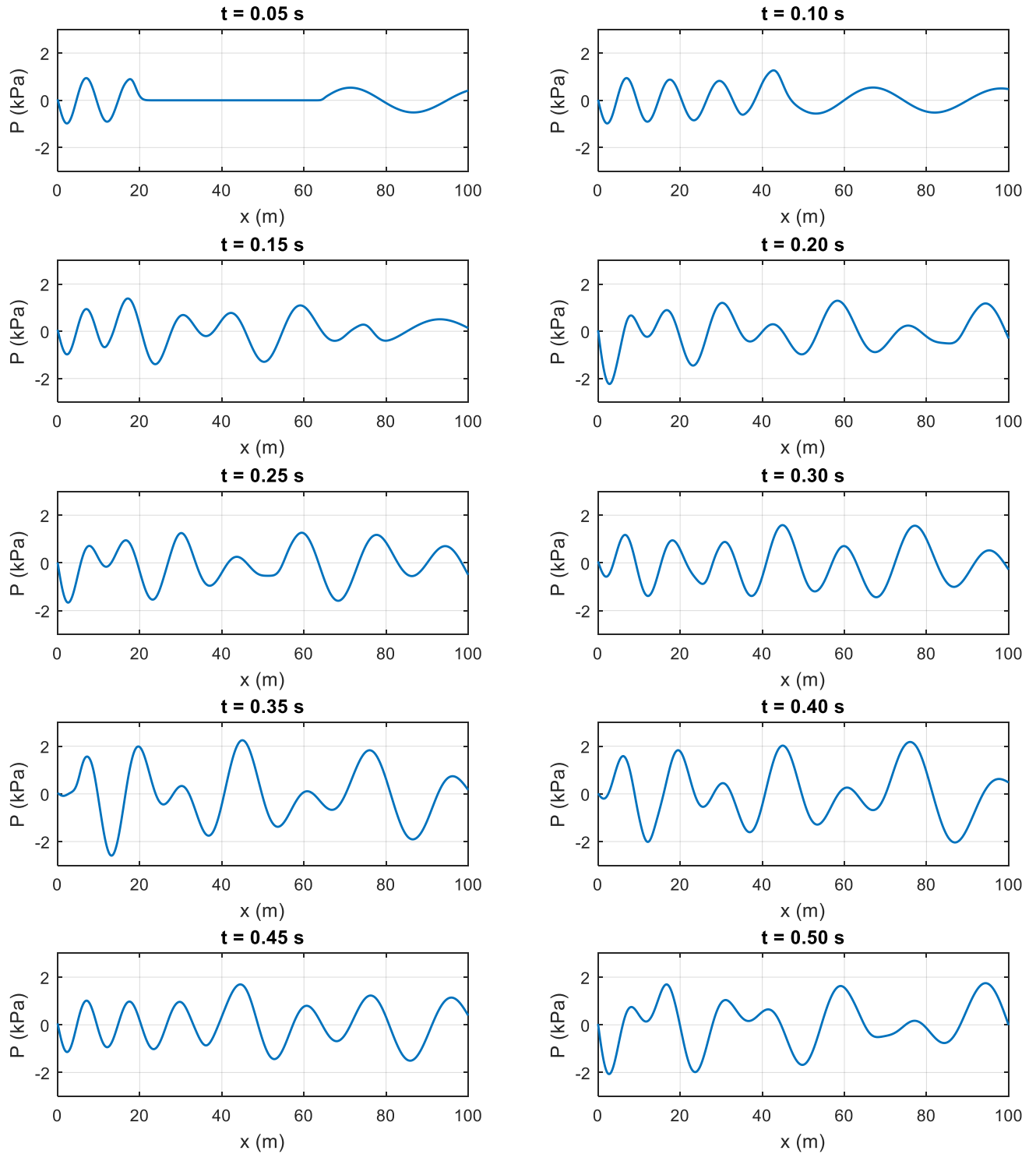


Fig 6. Pressure distribution time slices for fixed space step (h) and increasing T_0 ; $N=500$; pressure perturbation waves emanate from both boundaries harmonically with a different amplitude and frequency at each boundary.

Figure 6 illustrates the character of the wave interference and reflection off the boundaries for a prolonged duration.

7.3 Variable space step (h) and increasing T_o

Having an $a_o k/h$ that cannot be set equal 1.0 at all nodes for increasing T_o with constant h is an issue since we desire to have an exact solution, which requires that $a_o k/h=1$. To solve this issue, a Lax-Wendroff scheme was derived to accommodate a variable h . To maintain $a_o k/h \approx 1$ with constant k , we can vary h in accordance with the increasing a_o in the x -direction. It turns out to be a non-trivial exercise since a_o increases non-linearly. A solution for the h values was not found analytically, even after significant effort; however, an excel solver proved very useful for the purpose. Supporting calculations and solver results for h are presented in Appendix III.

Given that h is no longer constant, new centered finite difference equations were required to handle different values for h on either side of a given node. These equations, for generic first and second partial derivatives respectively, were derived in a previous course using method of undetermined coefficients and are:

$$\frac{\partial f_j^n}{\partial x} = -\frac{h_r}{h_l(h_l + h_r)} f_{j-1}^n + \frac{h_r - h_l}{h_l h_r} f_j^n + \frac{h_l}{h_r(h_l + h_r)} f_{j+1}^n \quad (33)$$

$$\frac{\partial^2 f_j^n}{\partial x^2} = \frac{2}{h_l(h_l + h_r)} f_{j-1}^n - \frac{2}{h_l h_r} f_j^n + \frac{2}{h_r(h_l + h_r)} f_{j+1}^n \quad (34)$$

Equations 33 and 34 are not second order accurate due to the different h values; however, since the leading truncation error contains the difference between h_l and h_r , which is small for adjacent nodes for the case studied, these equations do approach second order. These equations replace the existing space derivatives in the Lax-Wendroff scheme. The updated equations are shown in Appendix IV. The following table presents the parameters for the tested case, which is similar Case 4 (constant h), but for variable h .

Table 3. Parameters for variable space step (h) and increasing T_o .

		Case
Parameter	Units	5 (Fig 7)
Pipe length (L)	m	100
Number of nodes (incl. phantom) (N)	d-less	200
x-Dimension spacing (h)	m	0.505
Time step (k)	s	0.000897
$a_o k/h$	d-less	1.0
Duration	s	0.5
T_o	K	300 at LHS, linear to 1500 at RHS
<i>Boundary condition LHS</i>		
Perturbation pressure amplitude (A_{P_b})	Pa	1000
Frequency (f)	Hz	40
<i>Boundary condition RHS</i>		
Perturbation pressure amplitude (A_{P_b})	Pa	500
Frequency (f)	Hz	23

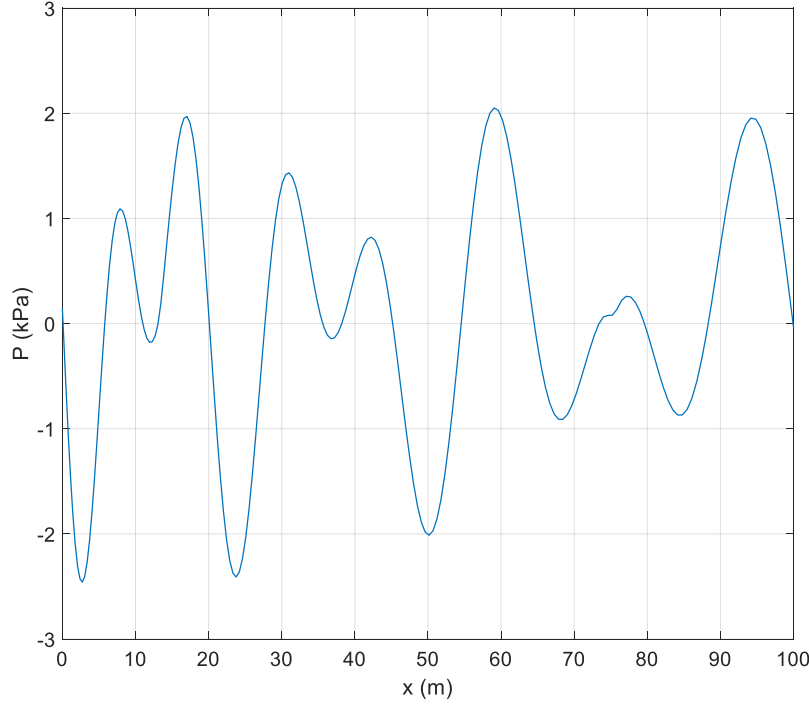


Fig 7. Pressure distribution for variable space step (h) and increasing T_o for 0.5s duration with $N=200$.

Figure 7 (Case 5) is a direct comparison to the last plot in figure 6 (Case 4), since all parameters in both cases are the same except for h , which is assumed to be constant for Case 4 and variable for Case 5, and N , which for Case 4 is 500 and Case 5 is 200. The character of the perturbation pressure is very similar for both cases, except that for Case 5 the peak pressures are slightly larger. It is intriguing to compare Case 5 to a case like Case 4 but for $N=200$, the same as Case 5. This makes for an even more direct comparison, with only the assumption for h different in both cases. Figure 8 compares the two cases:

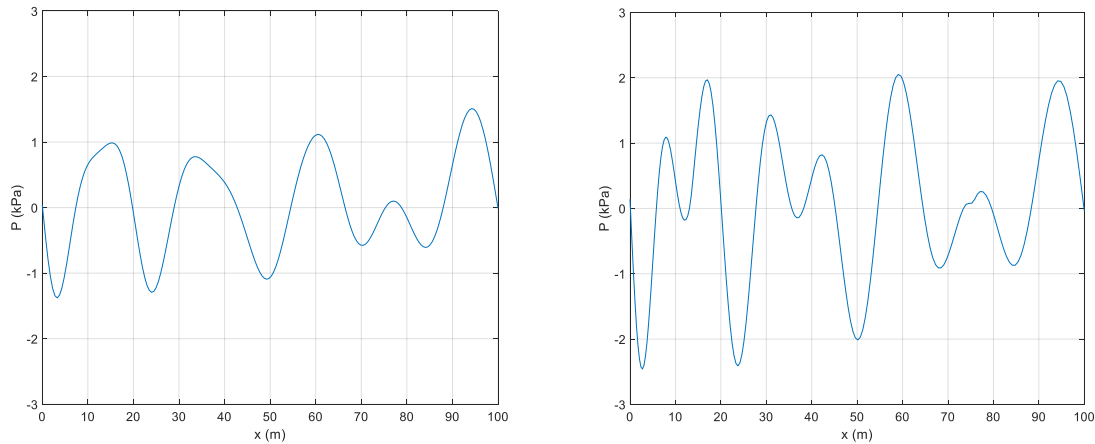


Fig 8. Comparison of Case 4 (constant h , $N=200$) on LHS, and Case 5 (variable h , $N=200$) on RHS.

The character of the pressure between the two cases shown in Fig 8 is vastly different. For constant h , the dispersion error is significant. This proves that using a variable h yields a significantly improved result over constant h . The variable h case with $N=200$ (Fig. 7 or RHS Fig. 8) yields a similar accuracy to the constant h case with N between 500 and 1000 (last plot Fig. 6). Simulation run times are significantly lower for variable h as well. The variable h case (case 5) for $N=200$ runs in 10.7 s, whereas for the similar accuracy case for constant h (Case 4) the run time is between 36 and 73 s.

8. Conclusions

The study confirms the practicality of the Lax-Wendroff one-step numerical scheme in solving the 1-D wave equation for a simulated pipe with a temperature profile and pistons providing harmonic pressure perturbations at either end. A modified Lax-Wendroff finite difference scheme was derived for the case of variable space step size (h). Several scenarios are presented, and results suggest that for the case of a linearly increasing temperature profile in the pipe, there is an advantage in terms of computation time and accuracy to varying h such that $a_o k/h = 1$, or just slightly less than 1, everywhere on the mesh.

9. Bibliography

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- [3] J. John D. Anderson, Computational Fluid Dynamics, USA: McGraw-Hill, Inc., 1995.

Appendix I

Nomenclature

Latin symbols

A_p	perturbation pressure amplitude (Pa)
a	speed of sound (m/s)
c	speed of wave (m/s)
h	space step size (m)
LHS	left hand side
l	left side
k	time step size (s)
P	pressure (Pa)
R	gas constant for air (kJ/kg·K)
RHS	right hand side
r	right side
T	temperature (K)
t	time (s)
u	wave particle velocity (m/s)
x	position (m)

Greek symbols

γ	specific heats ratio (d-less)
ρ	density (kg/m ³)
ω	harmonic frequency (rad/s)

Other superscripts, and subscripts

o	unperturbed state
$'$	perturbed state

Appendix II

MATLAB Scripts

```
% Pressure waves (planar) in a long pipe
% Boundary conditions: harmonic perturbation pressure at each end of pipe
% Initial conditions: pressure profile at t=0
% 2nd order linear hyperbolic equation (wave eqn)
% Lax-Wendroff scheme for system of 2 1st order linear hyp. eqns

clc
close all
clear variables
format compact

tic

global L To Po R gamma N;

% CONSTANT VARIABLES
t_max=0.5; % desired duration
N=200; % # nodes
L=100; % length of tube (m)
Po=101325; % unperturbed static pressure (Pa)
To=[300 1500]; % unperturbed static temp. at LHS and RHS of tube (K)
lambda_max=1.0; % desired lambda (max, since lambda may vary with x)
R=287.1; % gas constant for air (J/kg·K)
gamma=1.4; % assume constant over temperature range (D-less)
use_constant_h=0; % flag whether to use constant or variable h

% Initial condition values for perturbation pressure (P) and velocity (u)
P=ones(N,1)*0; % at t=0
% P(81:100,1)=100:100:2000;
% P(101:120,1)=1900:-100:0;
% P(75:126,1)=2000;
u=ones(N,1)*0; % at t=0

% Boundary condition (BC) parameters (for pistons at ends of pipe)
A=[1000 500]; % amplitude of piston pressure (LHS and RHS) (Pa)
f=[40 23]; % desired piston garmonic cycle frequency (LHS and RHS) (Hz)

% CALCULATED VARIABLES
% array for plotting solution at time intervals
time_snapshots = [0.05 0.1 0.15 0.2 0.25 0.3 0.35 0.4 0.45 0.5];
if use_constant_h == true
    h = L/(N-2); % space step (reflects 1 phantom node at each end)
    x=(h*((2:(N-1))-1.5))'; % create array of x values for nodes 2 to N-1
else
    Get_h(); % test whether ok to load array of varying h values from file
    load('h.mat'); % load array of varying h values from file
    x = Get_x(h); % create array of x values for nodes 2 to N-1
end
x=[0; x; L]; % add the end values (piston face) of 0 and L to x (nodes 1, N)
T=Temp(x); % create temp. array
row=Density(T); % create density array
c=SpeedOfSound(T); % create speed of sound array
% determine optimal time step for lambda=1 (s)
if use_constant_h == true
```



```

k=lambda_max*h/max(c);
else
k=h(1)/c(2);
end
w=2*pi*f; % piston frequency (LHS and RHS) (rad/s)

i = 1; % iterator for plot series
q = 1; % iterator for plot series
figure % start a figure for plotting
t_lapsed = 0; % elapsed time
while t_lapsed < t_max
    t_lapsed = t_lapsed + k;
    Pbound=A.*sin(w.*t_lapsed); % piston BC's
    dPdt=A.*w.*cos(w.*t_lapsed); % piston BC's
    P_old=P;
    u_old=u;
    for j=2:N-1
        if use_constant_h == true
            u(j)=u_old(j)-k/2/row(j)/h*(P_old(j+1)-P_old(j-1))+...
                (k*c(j)/h)^2/2*(u_old(j-1)-2*u_old(j)+u_old(j+1));
            P(j)=P_old(j)-k*gamma*Po/2/h*(u_old(j+1)-u_old(j-1))+...
                (k*c(j)/h)^2/2*(P_old(j-1)-2*P_old(j)+P_old(j+1));
        else
            u(j)=u_old(j)-k/row(j)*(h(j-1)/(h(j)*(h(j-1)+h(j)))*P_old(j+1)+...
                (h(j)-h(j-1))/(h(j-1)*h(j))*P_old(j)-...
                h(j)/(h(j-1)*(h(j-1)+h(j)))*P_old(j-1))+...
                k^2*c(j)^2/2*(2/(h(j-1)*(h(j-1)+h(j)))*u_old(j-1)-...
                2/(h(j-1)*h(j))*u_old(j)+...
                2/(h(j)*(h(j-1)+h(j)))*u_old(j+1));
            P(j)=P_old(j)-k*gamma*Po*(h(j-1)/(h(j)*(h(j-1)+h(j)))*u_old(j+1)+...
                (h(j)-h(j-1))/(h(j-1)*h(j))*u_old(j)-...
                h(j)/(h(j-1)*(h(j-1)+h(j)))*u_old(j-1))+...
                k^2*c(j)^2/2*(2/(h(j-1)*(h(j-1)+h(j)))*P_old(j-1)-...
                2/(h(j-1)*h(j))*P_old(j)+...
                2/(h(j)*(h(j-1)+h(j)))*P_old(j+1));
        end
    end
    % update phantom node values (2nd order)
    % LHS
    P(1)=2*Pbound(1)-P(2);
    if use_constant_h == true
        u(1)=u(2)+h/gamma/Po*dPdt(1);
    else
        u(1)=u(2)+h(1)/gamma/Po*dPdt(1);
    end
    % RHS
    P(N)=2*Pbound(2)-P(N-1);
    if use_constant_h == true
        u(N)=u(N-1)-h/gamma/Po*dPdt(2);
    else
        u(N)=u(N-1)-h(N-1)/gamma/Po*dPdt(2);
    end
end

pause(0.01);
% plot series
if true
    Pa=(P(1)+P(2))/2; % press. at LHS boundary
    Pb=(P(N-1)+P(N))/2; % press. at RHS boundary
    plot_P=[Pa; (P(2:N-1)); Pb]/1000;

```

```

        plot(x,plot_P,'LineWidth',1.25);
        xlabel('x (m)');
        ylabel('P (kPa)');
        axis([-inf, inf, -3, 3]);
        ax=gca; ax.XGrid = 'on'; ax.YGrid = 'on';
    end

    if t_lapsed >= time_snapshots(q)
        record_P(:,q)=[Pa; (P(2:N-1)); Pb]/1000;
        q=q+1;
    end
    i=i+1;
end
q=1;
if true
    figure();
    for plot_t=time_snapshots
        subplot(5,2,q)
        plot(x,record_P(:,q),'LineWidth',1.25);
        title_1 = sprintf('t = %0.2f s',plot_t);
        title(title_1);
        xlabel('x (m)');
        ylabel('P (kPa)');
        axis([-inf, inf, -3, 3]);
        ax=gca; ax.XGrid = 'on'; ax.YGrid = 'on';
        q=q+1;
    end
end
toc

```

```

function Get_h()
% tests whether or not it's ok to load array of varying h values from file
% parameters were calculated using solver in Excel and are specific to
% ONE SCENARIO ONLY:
% N=200
% L=100m
% TL=300k
% TR=1500k
% output: array of h values
global L To N;
if L ~= 100 || N ~= 200 || To(1) ~= 300 || To(2) ~= 1500
    error('Variable h applies to specific scenario only')
end

```

```

function x=Get_x(h)
% creates array of varying x values for the mesh
% only generates x for interior points
% input: array of h values
% output: array of x values
global N;
x=zeros(N-2,1); % allocate memory
x(1)=h(1)/2;
for i=2:(N-2)
    x(i)=x(i-1)+h(i);
end

```

```

function result=Temp(x)
% calculates temp. at a point along tube
% assumes linear profile, with temp. spec. at LHS and RHS of tube, respectively
% input: position along tube from LHS (m)

```

```

% global: L: length of tube
% output: temp. (K)
global L To;
result=(To(2)-To(1))/L*x+To(1);

```

```

function result=Density(T)
% density of air
% assumes air behaves as ideal gas
% input: temperature at nodes along tube from LHS (K)
% output: density (kg/m3)
global Po R;
result=Po/R./T;

```

```

function result=SpeedOfSound(T)
% speed of sound in air
% assumes air behaves as ideal gas
% input: temperature at nodes along tube from LHS (K)
% output: speed of sound (m/s)
global R gamma;
result=sqrt(gamma*R.*T);

```

```

% Generate characteristics plot
% 2nd order linear hyperbolic equation (wave eqn) with linear temp. profile
% case is for increasing To from 300K to 1500K an dL=100m

```

```

clc
close all
clear variables
format compact

```

```

a=1.4*287.1*300;
b=1.4*287.1*12;
figure();
for i=0:10
    x=i*10;
    xo=-2/b*sqrt(a+b*x);
    x=0:100;
    t=2/b*sqrt(a+b*x)+xo;
    plot(x,t,'k','LineWidth',1.25);
    hold on;
end

```

```

for i=0:10
    x=i*10;
    xo=2/b*sqrt(a+b*x);
    x=0:100;
    t=-2/b*sqrt(a+b*x)+xo;
    plot(x,t,'k','LineWidth',1.25);
    hold on;
end
% title({'Characteristics Trajectories','L=100'});
xlabel('x (m)')
ylabel('t (s)')
axis([-inf, inf, 0, inf]);
ax=gca; ax.XGrid = 'on'; ax.YGrid = 'on';

```

Appendix III

Solution of Variable Space Step Size

	Scen A					
N	200					
L	100					
TL	300					
TR	1500					
m	12					
k=	0.000897		6.50E-04	1.38E+00		
h num	h	x	T	a	a*k	a*k-h
		0				
1	0.31248	0.15624	301.8749	348.3326	0.31248	1.08E-10
2	0.314427	0.470667	305.648	350.5027	0.314427	4.33E-10
3	0.316374	0.787041	309.4445	352.6728	0.316374	4.33E-10
4	0.31832	1.105361	313.2643	354.8429	0.31832	4.33E-10
5	0.320267	1.425628	317.1075	357.0129	0.320267	4.33E-10
6	0.322214	1.747842	320.9741	359.1829	0.322214	4.33E-10
7	0.32416	2.072002	324.864	361.3528	0.32416	4.32E-10
8	0.326107	2.398108	328.7773	363.5227	0.326107	4.32E-10
9	0.328053	2.726162	332.7139	365.6926	0.328053	4.32E-10
10	0.33	3.056162	336.6739	367.8624	0.33	4.32E-10
11	0.331946	3.388108	340.6573	370.0322	0.331946	4.31E-10
12	0.333893	3.722	344.664	372.2019	0.333893	4.31E-10
13	0.335839	4.057839	348.6941	374.3716	0.335839	4.31E-10
14	0.337785	4.395625	352.7475	376.5413	0.337785	4.3E-10
15	0.339732	4.735357	356.8243	378.7109	0.339732	4.3E-10
16	0.341678	5.077035	360.9244	380.8805	0.341678	4.3E-10
17	0.343624	5.420659	365.0479	383.0501	0.343624	4.29E-10
18	0.34557	5.766229	369.1948	385.2196	0.34557	4.29E-10
19	0.347517	6.113746	373.365	387.3891	0.347517	4.29E-10
20	0.349463	6.463209	377.5585	389.5586	0.349463	4.28E-10
21	0.351409	6.814618	381.7754	391.728	0.351409	4.28E-10
22	0.353355	7.167973	386.0157	393.8974	0.353355	4.27E-10
23	0.355301	7.523274	390.2793	396.0667	0.355301	4.27E-10
24	0.357247	7.880521	394.5663	398.2361	0.357247	4.26E-10
25	0.359193	8.239715	398.8766	400.4054	0.359193	4.26E-10
26	0.361139	8.600854	403.2102	402.5746	0.361139	4.25E-10
27	0.363085	8.963939	407.5673	404.7439	0.363085	4.24E-10
28	0.365031	9.32897	411.9476	406.9131	0.365031	4.24E-10
29	0.366977	9.695947	416.3514	409.0822	0.366977	4.23E-10
30	0.368923	10.06487	420.7784	411.2514	0.368923	4.23E-10
31	0.370869	10.43574	425.2289	413.4205	0.370869	4.22E-10

32	0.372815	10.80855	429.7026	415.5896	0.372815	4.21E-10
33	0.37476	11.18331	434.1998	417.7586	0.37476	4.21E-10
34	0.376706	11.56002	438.7202	419.9276	0.376706	4.2E-10
35	0.378652	11.93867	443.2641	422.0966	0.378652	4.2E-10
36	0.380598	12.31927	447.8312	424.2656	0.380598	4.19E-10
37	0.382543	12.70181	452.4218	426.4345	0.382543	4.18E-10
38	0.384489	13.0863	457.0356	428.6034	0.384489	4.18E-10
39	0.386435	13.47274	461.6728	430.7723	0.386435	4.17E-10
40	0.38838	13.86112	466.3334	432.9412	0.38838	4.16E-10
41	0.390326	14.25144	471.0173	435.11	0.390326	4.16E-10
42	0.392271	14.64371	475.7246	437.2788	0.392271	4.15E-10
43	0.394217	15.03793	480.4552	439.4476	0.394217	4.14E-10
44	0.396163	15.43409	485.2091	441.6163	0.396163	4.13E-10
45	0.398108	15.8322	489.9864	443.785	0.398108	4.13E-10
46	0.400053	16.23226	494.7871	445.9537	0.400053	4.12E-10
47	0.401999	16.63425	499.6111	448.1224	0.401999	4.11E-10
48	0.403944	17.0382	504.4584	450.291	0.403944	4.11E-10
49	0.40589	17.44409	509.3291	452.4596	0.40589	4.1E-10
50	0.407835	17.85192	514.2231	454.6282	0.407835	4.09E-10
51	0.409781	18.2617	519.1405	456.7968	0.409781	4.08E-10
52	0.411726	18.67343	524.0812	458.9653	0.411726	4.08E-10
53	0.413671	19.0871	529.0452	461.1339	0.413671	4.07E-10
54	0.415617	19.50272	534.0326	463.3024	0.415617	4.06E-10
55	0.417562	19.92028	539.0434	465.4708	0.417562	4.05E-10
56	0.419507	20.33979	544.0774	467.6393	0.419507	4.05E-10
57	0.421452	20.76124	549.1349	469.8077	0.421452	4.04E-10
58	0.423397	21.18464	554.2156	471.9761	0.423397	4.03E-10
59	0.425343	21.60998	559.3197	474.1445	0.425343	4.02E-10
60	0.427288	22.03727	564.4472	476.3128	0.427288	4.01E-10
61	0.429233	22.4665	569.598	478.4812	0.429233	4.01E-10
62	0.431178	22.89768	574.7721	480.6495	0.431178	4E-10
63	0.433123	23.3308	579.9696	482.8178	0.433123	3.99E-10
64	0.435068	23.76587	585.1904	484.986	0.435068	3.98E-10
65	0.437013	24.20288	590.4346	487.1543	0.437013	3.98E-10
66	0.438959	24.64184	595.7021	489.3225	0.438959	3.97E-10
67	0.440904	25.08275	600.9929	491.4907	0.440904	3.96E-10
68	0.442849	25.52559	606.3071	493.6589	0.442849	3.95E-10
69	0.444794	25.97039	611.6446	495.827	0.444794	3.94E-10
70	0.446739	26.41713	617.0055	497.9952	0.446739	3.94E-10
71	0.448684	26.86581	622.3897	500.1633	0.448684	3.93E-10
72	0.450628	27.31644	627.7973	502.3314	0.450628	3.92E-10
73	0.452573	27.76901	633.2281	504.4995	0.452573	3.91E-10
74	0.454518	28.22353	638.6824	506.6675	0.454518	3.9E-10
75	0.456463	28.67999	644.1599	508.8356	0.456463	3.9E-10
76	0.458408	29.1384	649.6608	511.0036	0.458408	3.89E-10

77	0.460353	29.59875	655.185	513.1716	0.460353	3.88E-10
78	0.462298	30.06105	660.7326	515.3396	0.462298	3.87E-10
79	0.464243	30.52529	666.3035	517.5075	0.464243	3.86E-10
80	0.466187	30.99148	671.8978	519.6755	0.466187	3.85E-10
81	0.468132	31.45961	677.5154	521.8434	0.468132	3.85E-10
82	0.470077	31.92969	683.1563	524.0113	0.470077	3.84E-10
83	0.472022	32.40171	688.8205	526.1792	0.472022	3.83E-10
84	0.473966	32.87568	694.5081	528.347	0.473966	3.82E-10
85	0.475911	33.35159	700.2191	530.5149	0.475911	3.82E-10
86	0.477856	33.82945	705.9533	532.6827	0.477856	3.81E-10
87	0.479801	34.30925	711.711	534.8505	0.479801	3.8E-10
88	0.481745	34.79099	717.4919	537.0183	0.481745	3.79E-10
89	0.48369	35.27468	723.2962	539.1861	0.48369	3.78E-10
90	0.485635	35.76032	729.1238	541.3539	0.485635	3.78E-10
91	0.487579	36.24789	734.9747	543.5216	0.487579	3.77E-10
92	0.489524	36.73742	740.849	545.6893	0.489524	3.76E-10
93	0.491468	37.22889	746.7466	547.857	0.491468	3.75E-10
94	0.493413	37.7223	752.6676	550.0247	0.493413	3.74E-10
95	0.495357	38.21766	758.6119	552.1924	0.495357	3.74E-10
96	0.497302	38.71496	764.5795	554.3601	0.497302	3.73E-10
97	0.499247	39.21421	770.5705	556.5277	0.499247	3.72E-10
98	0.501191	39.7154	776.5848	558.6953	0.501191	3.71E-10
99	0.503136	40.21853	782.6224	560.8629	0.503136	3.7E-10
100	0.50508	40.72361	788.6834	563.0305	0.50508	3.7E-10
101	0.507025	41.23064	794.7676	565.1981	0.507025	3.69E-10
102	0.508969	41.73961	800.8753	567.3657	0.508969	3.68E-10
103	0.510913	42.25052	807.0062	569.5332	0.510913	3.67E-10
104	0.512858	42.76338	813.1605	571.7007	0.512858	3.66E-10
105	0.514802	43.27818	819.3382	573.8683	0.514802	3.66E-10
106	0.516747	43.79493	825.5391	576.0358	0.516747	3.65E-10
107	0.518691	44.31362	831.7634	578.2032	0.518691	3.64E-10
108	0.520635	44.83425	838.011	580.3707	0.520635	3.63E-10
109	0.52258	45.35683	844.282	582.5382	0.52258	3.63E-10
110	0.524524	45.88136	850.5763	584.7056	0.524524	3.62E-10
111	0.526469	46.40783	856.8939	586.873	0.526469	3.61E-10
112	0.528413	46.93624	863.2349	589.0404	0.528413	3.6E-10
113	0.530357	47.4666	869.5991	591.2078	0.530357	3.59E-10
114	0.532301	47.9989	875.9868	593.3752	0.532301	3.59E-10
115	0.534246	48.53314	882.3977	595.5426	0.534246	3.58E-10
116	0.53619	49.06933	888.832	597.7099	0.53619	3.57E-10
117	0.538134	49.60747	895.2896	599.8772	0.538134	3.56E-10
118	0.540079	50.14755	901.7706	602.0446	0.540079	3.56E-10
119	0.542023	50.68957	908.2748	604.2119	0.542023	3.55E-10
120	0.543967	51.23354	914.8024	606.3792	0.543967	3.54E-10
121	0.545911	51.77945	921.3534	608.5464	0.545911	3.53E-10

122	0.547855	52.3273	927.9276	610.7137	0.547855	3.53E-10
123	0.5498	52.8771	934.5252	612.881	0.5498	3.52E-10
124	0.551744	53.42885	941.1461	615.0482	0.551744	3.51E-10
125	0.553688	53.98253	947.7904	617.2154	0.553688	3.5E-10
126	0.555632	54.53817	954.458	619.3826	0.555632	3.49E-10
127	0.557576	55.09574	961.1489	621.5498	0.557576	3.49E-10
128	0.55952	55.65526	967.8631	623.717	0.55952	3.48E-10
129	0.561464	56.21673	974.6007	625.8842	0.561464	3.47E-10
130	0.563409	56.78014	981.3616	628.0513	0.563409	3.47E-10
131	0.565353	57.34549	988.1459	630.2185	0.565353	3.46E-10
132	0.567297	57.91278	994.9534	632.3856	0.567297	3.45E-10
133	0.569241	58.48203	1001.784	634.5527	0.569241	3.44E-10
134	0.571185	59.05321	1008.639	636.7199	0.571185	3.44E-10
135	0.573129	59.62634	1015.516	638.8869	0.573129	3.43E-10
136	0.575073	60.20141	1022.417	641.054	0.575073	3.42E-10
137	0.577017	60.77843	1029.341	643.2211	0.577017	3.41E-10
138	0.578961	61.35739	1036.289	645.3882	0.578961	3.41E-10
139	0.580905	61.93829	1043.26	647.5552	0.580905	3.4E-10
140	0.582849	62.52114	1050.254	649.7222	0.582849	3.39E-10
141	0.584793	63.10594	1057.271	651.8893	0.584793	3.39E-10
142	0.586737	63.69267	1064.312	654.0563	0.586737	3.38E-10
143	0.588681	64.28135	1071.376	656.2233	0.588681	3.37E-10
144	0.590625	64.87198	1078.464	658.3902	0.590625	3.36E-10
145	0.592569	65.46455	1085.575	660.5572	0.592569	3.36E-10
146	0.594513	66.05906	1092.709	662.7242	0.594513	3.35E-10
147	0.596457	66.65552	1099.866	664.8911	0.596457	3.34E-10
148	0.5984	67.25392	1107.047	667.0581	0.5984	3.34E-10
149	0.600344	67.85426	1114.251	669.225	0.600344	3.33E-10
150	0.602288	68.45655	1121.479	671.3919	0.602288	3.32E-10
151	0.604232	69.06078	1128.729	673.5588	0.604232	3.32E-10
152	0.606176	69.66696	1136.004	675.7257	0.606176	3.31E-10
153	0.60812	70.27508	1143.301	677.8926	0.60812	3.3E-10
154	0.610064	70.88514	1150.622	680.0595	0.610064	3.3E-10
155	0.612008	71.49715	1157.966	682.2263	0.612008	3.29E-10
156	0.613951	72.1111	1165.333	684.3932	0.613951	3.28E-10
157	0.615895	72.727	1172.724	686.56	0.615895	3.28E-10
158	0.617839	73.34484	1180.138	688.7269	0.617839	3.27E-10
159	0.619783	73.96462	1187.575	690.8937	0.619783	3.26E-10
160	0.621727	74.58634	1195.036	693.0605	0.621727	3.25E-10
161	0.62367	75.21001	1202.52	695.2273	0.62367	3.25E-10
162	0.625614	75.83563	1210.028	697.3941	0.625614	3.24E-10
163	0.627558	76.46319	1217.558	699.5608	0.627558	3.23E-10
164	0.629502	77.09269	1225.112	701.7276	0.629502	3.23E-10
165	0.631445	77.72413	1232.69	703.8944	0.631445	3.22E-10
166	0.633389	78.35752	1240.29	706.0611	0.633389	3.21E-10

167	0.635333	78.99286	1247.914	708.2278	0.635333	3.21E-10
168	0.637277	79.63013	1255.562	710.3946	0.637277	3.2E-10
169	0.63922	80.26935	1263.232	712.5613	0.63922	3.19E-10
170	0.641164	80.91052	1270.926	714.728	0.641164	3.19E-10
171	0.643108	81.55362	1278.643	716.8947	0.643108	3.18E-10
172	0.645051	82.19867	1286.384	719.0613	0.645051	3.17E-10
173	0.646995	82.84567	1294.148	721.228	0.646995	3.17E-10
174	0.648939	83.49461	1301.935	723.3947	0.648939	3.16E-10
175	0.650882	84.14549	1309.746	725.5613	0.650882	3.16E-10
176	0.652826	84.79832	1317.58	727.728	0.652826	3.15E-10
177	0.65477	85.45309	1325.437	729.8946	0.65477	3.14E-10
178	0.656713	86.1098	1333.318	732.0612	0.656713	3.14E-10
179	0.658657	86.76846	1341.221	734.2279	0.658657	3.13E-10
180	0.6606	87.42906	1349.149	736.3945	0.6606	3.12E-10
181	0.662544	88.0916	1357.099	738.5611	0.662544	3.12E-10
182	0.664488	88.75609	1365.073	740.7277	0.664488	3.11E-10
183	0.666431	89.42252	1373.07	742.8942	0.666431	3.1E-10
184	0.668375	90.09089	1381.091	745.0608	0.668375	3.1E-10
185	0.670318	90.76121	1389.135	747.2274	0.670318	3.09E-10
186	0.672262	91.43347	1397.202	749.3939	0.672262	3.08E-10
187	0.674205	92.10768	1405.292	751.5605	0.674205	3.08E-10
188	0.676149	92.78383	1413.406	753.727	0.676149	3.07E-10
189	0.678092	93.46192	1421.543	755.8935	0.678092	3.07E-10
190	0.680036	94.14196	1429.703	758.06	0.680036	3.06E-10
191	0.681979	94.82394	1437.887	760.2265	0.681979	3.05E-10
192	0.683923	95.50786	1446.094	762.393	0.683923	3.05E-10
193	0.685866	96.19373	1454.325	764.5595	0.685866	3.04E-10
194	0.68781	96.88154	1462.578	766.726	0.68781	3.03E-10
195	0.689753	97.57129	1470.855	768.8925	0.689753	3.03E-10
196	0.691697	98.26299	1479.156	771.0589	0.691697	3.02E-10
197	0.69364	98.95663	1487.48	773.2254	0.69364	3.02E-10
198	0.695584	99.65221	1495.827	775.3918	0.695584	3.01E-10
199	0.695584	100				

k adjustments

trial	k	end x	
1	0.000945	107.5031	
2	0.000944	107.3443	
3	0.0009	100.4518	
4	0.00089	98.91111	
5	0.000897	99.99901	0.000897
6	0.000897	100	0.000897
7			0.000897
8			

Appendix IV

Lax-Wendroff Scheme for Variable Space Step Size

Lax-Wendroff Scheme for Variable Space Step Size (h)

$$u_j^{n+1} = u_j^n - \frac{k}{\rho_0} \left(\frac{p_{j+1}^n - p_{j-1}^n}{2h} \right) + \frac{k^2 a^2}{2} \left(\frac{u_{j-1}^n - 2u_j^n + u_{j+1}^n}{h^2} \right)$$

$$u_j^{n+1} = u_j^n - \frac{k}{\rho_0} \left[\frac{h_1}{h_2(h_1+h_2)} p_{j+1}^n + \frac{h_2-h_1}{h_1 h_2} p_j^n - \frac{h_2}{h_1(h_1+h_2)} p_{j-1}^n \right] + \frac{k^2 a^2}{2} \left[\frac{2}{h_1(h_1+h_2)} u_{j-1}^n - \frac{2}{h_1 h_2} u_j^n + \frac{2}{h_2(h_1+h_2)} u_{j+1}^n \right]$$

$$p_j^{n+1} = p_j^n - k \gamma \rho_0 \left[\frac{u_{j+1}^n - u_{j-1}^n}{2h} \right] + \frac{k^2 a^2}{2} \left[\frac{p_{j-1}^n - 2p_j^n + p_{j+1}^n}{h^2} \right]$$

$$p_j^{n+1} = p_j^n - k \gamma \rho_0 \left[\frac{h_1}{h_2(h_1+h_2)} u_{j+1}^n + \frac{h_2-h_1}{h_1 h_2} u_j^n - \frac{h_2}{h_1(h_1+h_2)} u_{j-1}^n \right] + \frac{k^2 a^2}{2} \left[\frac{2}{h_1(h_1+h_2)} p_{j-1}^n - \frac{2}{h_1 h_2} p_j^n + \frac{2}{h_2(h_1+h_2)} p_{j+1}^n \right]$$